

3 The photograph shows a gentoo penguin from Antarctica.



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These penguins swim in the very cold Antarctic waters in search of food.

They feed on fish, squid and shrimp.

- (a) Explain **two** anatomical adaptations of these penguins that enable them to survive in this environment.

Use the information in the question to support your answer.

(2)

Adaptation 1

Reason

.....

.....

Adaptation 2

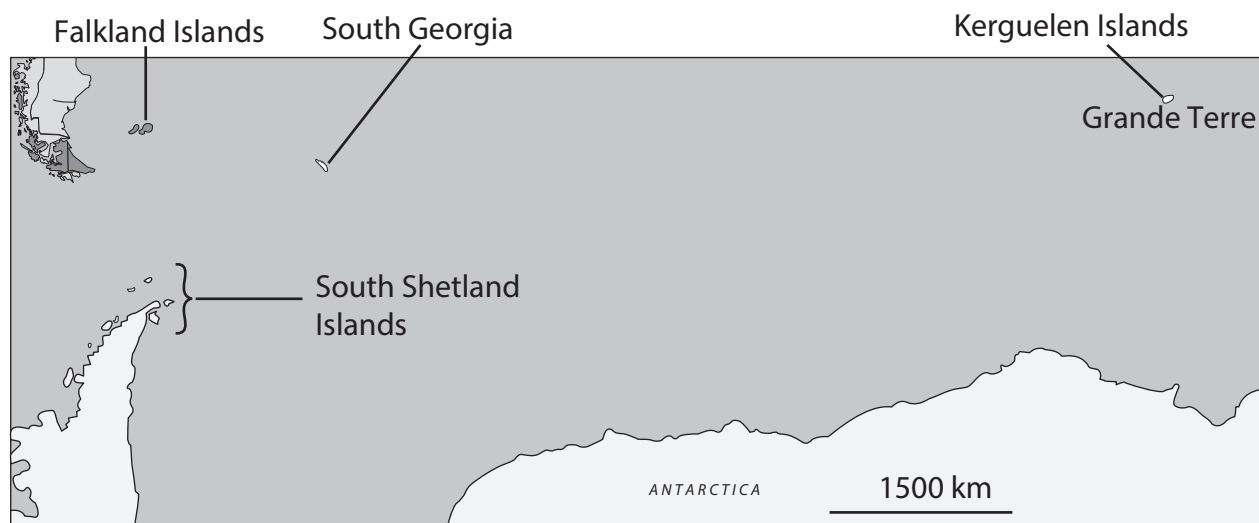
Reason

.....

.....



(b) The map shows four groups of islands that have gentoo penguin populations.



It has been proposed that the gentoo penguins on these four groups of islands are **different** species.

- (i) Suggest **one** way that these penguins can be shown to belong to different species.

(1)

(ii) Explain how different species of gentoo penguin could have formed.

Use the map to support your answer.

(5)

(Total for Question 3 = 8 marks)

